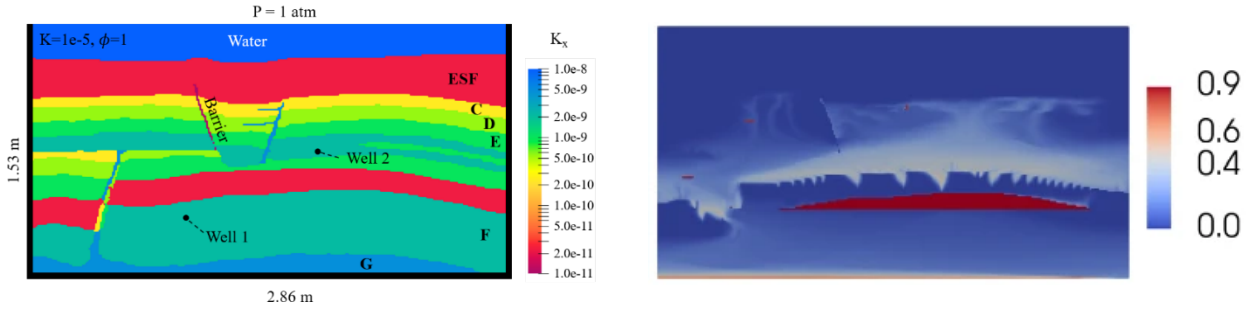
DARSim Example:

FluidFlower benchmark



This example presents the FluidFlower benchmark setup implemented in DARSim. The figures show the problem geometry and permeability distribution (left) and the dissolved CO_2 distribution, represented by (R_s), after five days (right).

The problem setup involves a 2D box measuring 1.53 m in height, 2.86 m in length, and 1.9 cm in width, filled with seven distinct soil types to represent realistic geological formations, including a barrier and fractures. Two low-permeability layers, modeled as ESF soil, serve to mimic the effect of a cap rock. The top boundary of the box is filled with water maintained at a constant height and is open to flow under atmospheric pressure, which is modeled as a porous medium with porosity of $\phi = 1$ and a very high permeability of $k = 10^{-5}$ m^2. The other three boundaries are sealed as no-flow conditions. Two injection wells are used to inject pure gas at a constant rate of 1.44×10^{-4} m^3/s. The first well injects continuously from the start of the simulation for 5 hours, while the second well begins at 2.25 hours and continues for 2.75 hours.

The simulations are done under isothermal conditions of 20 ˚C and continue for 2 days. Further information is available in Flemisch et al. (2024), Wang et al. (2023), and Misaghi-Bonabi et al. (2025).